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*
*      STAAD.Pro
*      Version 2007 Build 04
*      Proprietary Program of
*      Research Engineers, Intl.
*      Date= FEB 28, 2013
*      Time= 16: 9:44
*
*      USER ID: Parsons Binckerhoff
*****

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1. STAAD SPACE
INPUT FILE: 130222 VM1 STAIRCASE 1.STD
2. START JOB INFORMATION
3. ENGINEER DATE 26-NOV-12
4. JOB NAME VM1 STAIRCASE
5. END JOB INFORMATION
6. INPUT WIDTH 79
7. UNIT METER KN
8. JOINT COORDINATES
9. 1 -107.085 0 -23.25; 2 -107.085 0 -22.05; 3 -107.085 0 -19.3
10. 4 -107.085 3 -23.25; 5 -107.085 3 -22.05; 6 -106.085 0 -23.25
11. 7 -106.085 0 -22.05; 8 -106.085 0 -19.3; 9 -106.085 3 -23.25
12. 10 -106.085 3 -22.05; 11 -103.085 0 -23.25; 12 -103.085 0 -22.05
13. 13 -103.085 3 -23.25; 14 -103.085 3 -22.05; 15 -103.085 4.8 -23.25
14. 16 -103.085 4.8 -22.05; 17 -103.085 5.8 -23.25; 18 -103.085 5.8 -22.05
15. 19 -102.085 5.8 -23.25; 20 -102.085 5.8 -22.05
16. MEMBER INCIDENCES
17. 1 4 1; 2 4 2; 3 5 1; 4 5 2; 5 5 3; 6 5 4; 7 9 4; 8 10 5; 9 9 6; 10 9 7
18. 11 10 6; 12 10 7; 13 10 8; 14 10 9; 15 17 9; 16 18 10; 17 13 11; 18 14 11
19. 19 13 12; 20 14 12; 21 14 13; 22 15 13; 23 16 14; 24 17 14; 25 18 13; 26 17 15
20. 27 18 16; 28 18 17; 29 19 15; 30 20 16; 31 17 19; 32 18 20; 33 20 19
21. DEFINE MATERIAL START
22. ISOTROPIC STEEL
23. E 2.05E+008
24. POISSON 0.3
25. DENSITY 76.8195
26. ALPHA 1.2E-005
27. DAMP 0.03
28. END DEFINE MATERIAL
29. MEMBER PROPERTY JAPANESE
30. 17 20 22 23 26 27 TABLE ST H250X250X9
31. 1 4 9 12 TABLE ST H250X250X9
32. 7 8 31 32 TABLE ST H250X250X9
33. 5 13 15 16 TABLE ST C250X90X9
34. 2 3 10 11 18 19 24 25 TABLE ST C150X75X6.5
35. 6 14 21 28 TO 30 33 TABLE ST H200X200X8
36. CONSTANTS
37. MATERIAL STEEL ALL
38. BETA 180 MEMB 5 16
39. SUPPORTS
40. 1 TO 3 6 TO 8 PINNED
41. 11 12 FIXED
42. MEMBER RELEASE
43. 15 16 END MY MZ
44. 5 13 15 16 START MY MZ
45. 28 33 START MY MZ
46. 21 START MY MZ
47. 21 END MY MZ
48. 28 33 END MY MZ
49. 14 START MY MZ
50. 14 END MY MZ
51. 6 START MY MZ
52. 6 END MY MZ
53. MEMBER TENSION

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54. 18 19
55. MEMBER TENSION
56. 18 19
57. MEMBER TENSION
58. 2 3 10 11
59. MEMBER TENSION
60. 24 25
61. LOAD 1 LOADTYPE NONE TITLE DL
62. SELFWEIGHT Y -1 LIST 1 TO 33
63. MEMBER LOAD
64. *0.75KPA DEAD LOAD*
65. 31 32 UNI GY -0.25
66. 5 7 8 13 15 16 UNI GY -0.25
67. LOAD 2 LOADTYPE NONE TITLE LL
68. *5KPA LIVE LOAD*
69. MEMBER LOAD
70. 31 32 UNI GY -2.5
71. 5 7 8 13 15 16 UNI GY -2.5
72. LOAD 3 LOADTYPE NONE TITLE WLX
73. MEMBER LOAD
74. 2 3 10 11 18 19 24 25 UNI GX -0.15
75. 1 4 TO 6 9 12 TO 17 20 22 23 26 TO 28 33 UNI GX -0.24
76. LOAD 4 LOADTYPE NONE TITLE WLZ
77. MEMBER LOAD
78. 1 4 5 7 TO 9 12 13 15 TO 17 20 22 23 26 27 29 TO 32 UNI GZ -0.24
79. LOAD 5 LOADTYPE NONE TITLE NLX
80. SELFWEIGHT X -0.015 LIST 1 TO 33
81. MEMBER LOAD
82. 31 32 UNI GX -0.004
83. 5 7 8 13 15 16 UNI GX -0.004
84. LOAD 6 LOADTYPE NONE TITLE NLZ
85. SELFWEIGHT Z -0.015 LIST 1 TO 33
86. MEMBER LOAD
87. 31 32 UNI GZ -0.004
88. 5 7 8 13 15 16 UNI GZ -0.004
89. LOAD 7 LOADTYPE NONE TITLE 1.0DL+1.0LL
90. REPEAT LOAD
91. 1 1.0 2 1.0
92. LOAD 8 LOADTYPE NONE TITLE 1.0DL+1.0LL+1.0WLX
93. REPEAT LOAD
94. 1 1.0 2 1.0 3 1.0
95. LOAD 9 LOADTYPE NONE TITLE 1.0DL+1.0LL+1.0WLZ
96. REPEAT LOAD
97. 1 1.0 2 1.0 4 1.0
98. LOAD 10 LOADTYPE NONE TITLE 1.0DL+1.0LL+1.0NLX
99. REPEAT LOAD
100. 1 1.0 2 1.0 5 1.0
101. LOAD 11 LOADTYPE NONE TITLE 1.0DL+1.0LL+1.0NLZ
102. REPEAT LOAD
103. 1 1.0 2 1.0 6 1.0
104. LOAD 21 LOADTYPE NONE TITLE 1.0DL+1.0LL-1.0 WLX
105. REPEAT LOAD
106. 1 1.0 2 1.0 3 -1.0
107. LOAD 22 LOADTYPE NONE TITLE 1.0DL+1.0LL-1.0 WLZ
108. REPEAT LOAD
109. 4 -1.0 1 1.0 2 1.0
110. LOAD 23 LOADTYPE NONE TITLE 1.0DL+1.0LL-1.0 NLX
111. REPEAT LOAD
112. 1 1.0 2 1.0 5 -1.0
113. LOAD 24 LOADTYPE NONE TITLE 1.0DL+1.0LL-1.0 NLZ
114. REPEAT LOAD
115. 6 -1.0 1 1.0 2 1.0
116. *ULTIMATE LOAD COMBINATIONS
117. LOAD 25 LOADTYPE NONE TITLE 1.0DL+1.6LL
118. REPEAT LOAD
119. 1 1.0 2 1.6
120. LOAD 26 LOADTYPE NONE TITLE 1.4DL+1.6LL
121. REPEAT LOAD
122. 1 1.4 2 1.6
123. LOAD 27 LOADTYPE NONE TITLE 1.2DL+1.2LL+1.2WLX

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**START ITERATION NO.	2		
**NOTE-Tension/Compression converged after	2 iterations,	Case=	6
**START ITERATION NO.	2		
**START ITERATION NO.	3		
**NOTE-Tension/Compression converged after	3 iterations,	Case=	7
**START ITERATION NO.	2		
**START ITERATION NO.	3		
**NOTE-Tension/Compression converged after	3 iterations,	Case=	8
**START ITERATION NO.	2		
**START ITERATION NO.	3		
**NOTE-Tension/Compression converged after	3 iterations,	Case=	9

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**START ITERATION NO.      2

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**NOTE-Tension/Compression converged after 3 iterations, Case= 10

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**START ITERATION NO.          2
**START ITERATION NO.          3

**NOTE-Tension/Compression converged after 3 iterations, Case= 11

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**START ITERATION NO.          2
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**START ITERATION NO.          2
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**START ITERATION NO.      2
**START ITERATION NO.      3

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**START ITERATION NO.          2
```

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**START ITERATION NO.      2
**START ITERATION NO.      3

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**NOTE-Tension/Compression converged after 3 iterations, Case= 25

**START ITERATION NO. 2
**START ITERATION NO. 3

**NOTE-Tension/Compression converged after 3 iterations, Case= 26

**START ITERATION NO. 2
**START ITERATION NO. 3

**NOTE-Tension/Compression converged after 3 iterations, Case= 27

**START ITERATION NO. 2
**START ITERATION NO. 3

**NOTE-Tension/Compression converged after 3 iterations, Case= 28

**START ITERATION NO. 2
**START ITERATION NO. 3

**NOTE-Tension/Compression converged after 3 iterations, Case= 29

**START ITERATION NO. 2
**START ITERATION NO. 3

**NOTE-Tension/Compression converged after 3 iterations, Case= 30

**START ITERATION NO. 2

**NOTE-Tension/Compression converged after 2 iterations, Case= 41

**START ITERATION NO. 2

**NOTE-Tension/Compression converged after 2 iterations, Case= 42

**START ITERATION NO. 2
**START ITERATION NO. 3

**NOTE-Tension/Compression converged after 3 iterations, Case= 43

**START ITERATION NO. 2
**START ITERATION NO. 3

**NOTE-Tension/Compression converged after 3 iterations, Case= 44

148. DEFINE ENVELOP
149. 25 TO 30 41 TO 44 ENVELOP 1 TYPE STRESS
150. 7 TO 11 21 TO 24 ENVELOP 2 TYPE SERVICEABILITY
151. END DEFINE ENVELOP
152. LOAD LIST 1 TO 11 21 TO 25

153. *PRINT JOINT DISPLACEMENTS ALL
154. LOAD LIST 26 TO 30 41 TO 44
155. *PRINT MEMBER FORCES ALL
156. LOAD LIST ALL
157. PARAMETER 1
158. CODE BS5950
159. SGR 0 ALL
160. *TRACK 2
161. CHECK CODE ALL

STAAD.Pro CODE CHECKING - (BSI)

PROGRAM CODE REVISION V2.12_5950-1_2000

ALL UNITS ARE - KN METE (UNLESS OTHERWISE NOTED)

MEMBER	TABLE	RESULT/ FX	CRITICAL COND/ MY	RATIO/ MZ	LOADING/ LOCATION
=====					
1 ST	H250X250X9	PASS	BS-4.8.2.2	0.030	41
		9.68 T	0.00	6.78	0.00
2 ST	C150X75X6.5	PASS	BS-4.6 (T)	0.003	41
		1.76 T	0.00	0.00	0.00
3 ST	C150X75X6.5	PASS	BS-4.6 (T)	0.003	41
		1.68 T	0.00	0.00	0.00
4 ST	H250X250X9	PASS	BS-4.8.2.2	0.034	41
		5.42 T	0.00	8.37	0.00
5 ST	C250X90X9	PASS	BS-4.8.3.3.1	0.172	41
		0.04 C	0.60	5.18	-
6 ST	H200X200X8	PASS	BS-4.2.3-(Y)	0.002	26
		0.16 C	0.00	0.00	0.00
7 ST	H250X250X9	PASS	BS-4.2.3-(Y)	0.043	41
		1.36 T	0.00	7.01	0.00
8 ST	H250X250X9	PASS	BS-4.2.3-(Y)	0.051	41
		1.31 T	0.00	8.59	0.00
9 ST	H250X250X9	PASS	BS-4.3.6	0.027	41
		22.10 C	0.00	7.01	0.00
10 ST	C150X75X6.5	PASS	BS-4.6 (T)	0.002	4
		1.15 T	0.00	0.00	0.00
11 ST	C150X75X6.5	PASS	BS-4.6 (T)	0.004	3
		2.48 T	0.00	0.00	0.00
12 ST	H250X250X9	PASS	BS-4.3.6	0.034	41
		32.92 C	0.00	8.60	0.00
13 ST	C250X90X9	PASS	BS-4.8.3.3.1	0.172	27
		0.04 C	0.60	5.18	-
14 ST	H200X200X8	PASS	BS-4.2.3-(Y)	0.002	26
		0.00	0.00	0.00	0.00
15 ST	C250X90X9	PASS	BS-4.8.3.3.1	0.180	28
		0.02 C	0.61	5.70	-
16 ST	C250X90X9	PASS	BS-4.8.3.3.1	0.180	42
		0.02 C	0.61	5.70	-
17 ST	H250X250X9	PASS	BS-4.3.6	0.020	27
		16.12 C	0.00	5.18	0.00
18 ST	C150X75X6.5	PASS	BS-4.6 (T)	0.018	42
		11.79 T	0.00	0.00	0.00
19 ST	C150X75X6.5	PASS	BS-4.6 (T)	0.018	28
		11.79 T	0.00	0.00	0.00
20 ST	H250X250X9	PASS	BS-4.3.6	0.023	27
		16.23 C	0.00	5.87	0.00
21 ST	H200X200X8	PASS	BS-4.7 (C)	0.002	28
		3.76 C	0.00	0.00	0.00
22 ST	H250X250X9	PASS	ANNEX I.1	0.011	28
		24.18 C	0.35	2.22	-

ALL UNITS ARE - KN METE (UNLESS OTHERWISE NOTED)

MEMBER	TABLE	RESULT/ FX	CRITICAL MY	COND/ MZ	RATIO/ MZ	LOADING/ LOCATION
23	ST	H250X250X9	PASS	ANNEX I.1	0.011	42
		24.18 C		0.35	2.22	-
24	ST	C150X75X6.5	PASS	BS-4.6 (T)	0.013	28
		8.61 T		0.00	0.00	0.00
25	ST	C150X75X6.5	PASS	BS-4.6 (T)	0.013	42
		8.61 T		0.00	0.00	0.00
26	ST	H250X250X9	PASS	ANNEX I.1	0.011	42
		11.63 C		0.63	1.41	-
27	ST	H250X250X9	PASS	ANNEX I.1	0.011	28
		11.63 C		0.63	1.40	-
28	ST	H200X200X8	PASS	BS-4.2.3-(Y)	0.002	26
		0.00		0.00	0.00	0.00
29	ST	H200X200X8	PASS	ANNEX I.1	0.012	28
		3.93 C		0.40	0.82	-
30	ST	H200X200X8	PASS	ANNEX I.1	0.012	42
		3.93 C		0.40	0.82	-
31	ST	H250X250X9	PASS	BS-4.2.3-(Y)	0.008	26
		2.41 T		0.00	-0.36	1.00
32	ST	H250X250X9	PASS	BS-4.2.3-(Y)	0.008	26
		2.41 T		0.00	-0.36	1.00
33	ST	H200X200X8	PASS	BS-4.2.3-(Y)	0.002	26
		0.00		0.00	0.00	0.00

***** END OF TABULATED RESULT OF DESIGN *****

162. FINISH

***** END OF THE STAAD.Pro RUN *****

**** DATE= FEB 28,2013 TIME= 16: 9:49 ****

* For questions on STAAD.Pro, please contact *

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* *

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